

Hüseyin Karaca

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🔗 huseyin-karaca.github.io in huseyinkaraca 🎓 Hüseyin Karaca 🌐 huseyin-karaca

Education

Master of Science in Electrical and Electronics Engineering <i>Bilkent University</i> Grade: 3.93/4.00 Advisor: Prof. Dr. Süleyman Serdar Kozat Thesis topic: Ensemble learning and feature selection methods for time-series forecasting	2024-07 – present Ankara, Türkiye
Bachelor of Science in Electrical and Electronics Engineering <i>Middle East Technical University</i> Grade: 3.60/4.00 Advisor: Assoc. Prof. Dr. Elif Vural Capstone Project: A smart monitor to keep track of vitals and sleep patterns of the baby.	2019-09 – 2024-07 Ankara, Türkiye
High School Diploma in Science <i>Ankara Science High School</i> Grade: 97.19/100	2015-09 – 2019-06 Ankara, Türkiye

Research Experience

Summer Intern <i>UCLA (The Ozcan Research Group)</i> Studied diffusion models for biomedical images under the supervision of Prof. Dr. Aydogan Ozcan. Helped the preparation of 3D printed diffractive layers for an all-optical image denoiser network.	2023-07 – 2023-09 Los Angeles, CA, USA
Undergraduate Researcher <i>METU (Dr. Elif Vural's Research Group)</i> Studied sample complexity of domain adaptation of different deep learning architectures. Performed regular neural network trainings with PyTorch to test novel theoretical bounds.	2022-09 – 2024-07 Ankara, Türkiye
Summer Intern <i>TUBITAK SAGE (Image Processing Division)</i> Studied on bad pixel detection and wrote a MATLAB code that simplifies the examination of calibration images. Improved robustness analysis algorithms by performing stitching on real seeker images.	2022-08 – 2022-09 Ankara, Türkiye
Undergraduate Researcher <i>METU (Communication Networks Group)</i> Studied age-aware random access, specifically LoRa - LoRaWAN protocols. Employed a LoRaWAN network at the METU campus.	2021-07 – 2022-09 Ankara, Türkiye

Publications

- J3 **Karaca, H.**, Tumay, A., Koc, A.T., Kozat, S.S.. “Hierarchical ensemble-based feature selection for time series prediction”. *Digital Signal Processing*, 2026. DOI: 10.1016/j.dsp.2025.105528
- J2 **Karaca, H.**, Kozat, S.S.. “Soft Gradient Boosting with Learnable Feature Transforms for Sequential Regression”. *arXiv preprint*, 2024. DOI: 10.48550/arXiv.2509.12920
- J1 Isil, C., Gan, T., Ardic, F.O., Montesoglu, K., Digani, J., **Karaca, H.**, Chen, H., Li, J., Mengü, D., Jarrahi, M., Aksit, K., Ozcan, A.. “All-optical image denoising using a diffractive visual processor”. *Light: Science & Applications*, 2024. DOI: 10.1038/s41377-024-01385-6
- C1 **Karaca, H.**, Akgul, O., Arslan, O.F., Aydemir, A.C., Aydin, F.S., Unsal, E.A., Vural, E.. “An Experimental Study of the Sample Complexity of Domain Adaptation”. 2023. DOI: 10.1109/SIU59756.2023.10223790
- P1 Vural, E., **Karaca, H.**. “A Unified Analysis of Generalization and Sample Complexity for Semi-Supervised Domain Adaptation”. *arXiv preprint*, 2025. DOI: 10.48550/arXiv.2507.22632

Teaching Experience

Teaching Assistant

Bilkent University

Course: EEE321 Signals and Systems

Guiding and evaluating students during laboratory sessions, grading reports and quizzes.
Organizing recitations to reinforce core lecture topics.

2024-07 – present

Ankara, Türkiye

Student Assistant

Middle East Technical University

Courses: EE201 Circuit Theory I EE224 Electromagnetic Theory

Organized weekly recitations and assisted the preparation of lecture notes.

2023-10 – 2024-06

Ankara, Türkiye

Skills

Languages: Turkish (Native), English (C1), Persian (A2)
Programming: PyTorch, Python, MATLAB, LaTeX, Scikit-learn, Nixtla, Sktime
Software: Siemens NX, LTspice, Wordpress, Quartus II, Modelsim, Simulink
Hardware: Oscilloscope, FPGA, SDR

Honors and Awards

TUBITAK Grants

2019-2024

2210-A: National Graduate Scholarship Program (2024)

2224-D: Support Program for Participation in Scientific Activities Abroad (2023)

2205: Undergraduate Scholarship Program (2019)

National University Entrance Examination (YKS)

2019-07

Turkey

Ranked 24th among 2.6M participants (Score: 482/500)

Awarded the Presidential Merit Scholarship and Is Bank Golden Youth Award.

TUBITAK National Physics Olympiads

2018-02

Turkey

Ranked 16th, won the bronze medal. Participated in national team selection bootcamp.

Extracurricular Activities

Trekking and Camping

2018-07

Participated in two trekking tours (200km) on the Ancient Lycian Way.

Organized midnight full moon walks and camps in Cappadocia.

METU Cycling Team

2021-10

Board Member

Represented METU in intercollegiate mountain and road bike races.

Student Council Presidency

2016-10

Ankara Science High School

Organized spring festivals and founded a French literature book society.